Dental Caries

Dental Caries

- **Calcified tissues**: Hard structures of the tooth, including enamel, dentin, and cementum.
- **Demineralization**: Loss of minerals, mainly calcium and phosphate, from the tooth structure.
- **Cavitation**: Formation of a cavity or hole in the tooth due to decay.

Classification of Caries

Morphology

- **Pit and fissure**: Grooves and depressions on the chewing surface of molars and premolars.
- **Smooth surface**: Flat surfaces of the teeth, such as the sides.

Dynamics

- **Acute**: Rapidly progressing decay.
- **Chronic**: Slow-progressing decay.

Nature of Attack

- **Primary (virgin) caries**: Decay occurring on a previously unaffected tooth surface.
- **Secondary (recurrent) caries**: Decay occurring around existing fillings or restorations.

Chronology

- **Infancy (nursing bottle) caries**: Decay in young children caused by prolonged bottle feeding with sugary liquids.
- **Adolescent caries**: Decay occurring commonly during teenage years.

Clinical Features of Dentinal Caries

- **Opaque**: Not transparent or clear.
- **Bluish-white**: A slight bluish discoloration seen on enamel.
- **Lateral spread**: Horizontal progression of decay.
- **DEJ (Dentinoenamel Junction)**: Boundary between enamel and dentin.

Histopathological Features of Dentinal Caries

- **Decalcification**: Loss of calcium salts from the tooth structure.
- **Proteolysis**: Breakdown of proteins, such as collagen, by enzymes.
- **Collagenous matrix**: Protein framework in dentin providing strength.

Zones of Dentinal Caries

Zone 1: Fatty Degeneration of Tome's Fibers

- **Tome's fibers**: Extensions of odontoblast cells into the dentinal tubules.
- **Intertubular dentin**: Dentin located between the tubules.
- **Apatite crystals**: Minerals that give dentin its hardness.

Zone 2: Dentinal Sclerosis

- **Odontoblastic processes**: Extensions of odontoblast cells within the dentinal tubules.
- **Remineralization**: Natural repair process where minerals are redeposited into demineralized enamel or dentin.

Zone 3: Decalcification of Dentin

- **Transparent**: Clear appearance of the decalcified zone.

Zone 4: Bacterial Invasion of Decalcified Dentin

- **Denaturation**: Breakdown of protein structure.
- **Cannot undergo self-repair**: Dentin in this zone is irreversibly damaged.

Zone 5: Decomposed Dentin

- **Liquefaction foci of Miller**: Areas where dentin has been destroyed by bacterial enzymes.
- **Transverse clefts**: Cracks running perpendicular to the dentinal tubules.
- **Cavitation**: Formation of a cavity due to complete destruction of dentin.

Theories of Dental Caries

Acidogenic Theory (Miller's Chemicoparasitic Theory)

- **Chemicoparasitic**: Refers to a chemical process where harmful microorganisms (parasites) play a role in causing damage to the tooth.
- **Decalcify**: Removal of calcium from tooth enamel and dentin.
- **Cariogenic**: Bacteria or substances that cause caries (tooth decay).
- **Cariogenic bacteria**: Bacteria responsible for the formation of dental caries.
- **Dextran**: A complex sugar produced by certain bacteria that aids in plaque formation.
- **Plaque**: A sticky film of bacteria that forms on the teeth.
- **Fermentable carbohydrates**: Sugars and starches that can be broken down by bacteria into acids.
- **pH**: A measure of acidity or alkalinity (low pH indicates acidity).
- **Neutralization**: The process of bringing the pH back to normal levels after it becomes acidic.

Proteolytic Theory

- **Proteolytic enzymes**: Enzymes that break down proteins, including those in the tooth enamel.
- **Organic matrix**: The protein-based structure within enamel that helps to hold the mineral content.
- **Inorganic crystals**: Mineral components like calcium and phosphate in enamel that provide hardness.
- **Proteolysis**: The breakdown of proteins into smaller fragments.

Proteolytic Chelation Theory

- **Chelation**: The process of binding metal ions (like calcium) with another substance, making them easier to remove.
- **Chelating agent**: A substance that binds to metal ions, such as calcium, and helps to remove them from the tooth.
- **Schatz et al**: Refers to the researchers who proposed this theory in 1955.

Sucrose Chelation Theory

- **Ionized calcium saccharate**: A complex formed when sucrose interacts with calcium, releasing ions.
- **Calcium saccharide**: A compound formed by the interaction of sucrose and calcium ions, leading to enamel demineralization.

Autoimmune Theory

- **Autoimmune mechanisms**: The process where the body's immune system mistakenly attacks its own cells or tissues.
- **Odontoblast cells**: Cells in the dental pulp responsible for forming dentin.
- **Pulp**: The soft tissue inside the tooth that contains blood vessels, nerves, and odontoblasts.

Histopathology of Enamel Caries

Early Enamel Caries

- **Interprismatic/inter-rod substances**: The materials between enamel rods that help hold them together.
- **Enamel rods**: The basic structural units of enamel, arranged in a parallel fashion.
- **Incremental striae of Retzius**: Microscopic lines or bands in enamel that represent periods of enamel formation.
- **Perikymata**: Surface markings or grooves corresponding to the striae of Retzius, visible on the tooth's surface.

Advanced Enamel Caries

- **Translucent zone**: The deepest part of the carious lesion, where the enamel is more porous and appears clearer or less opaque.
- **Pores**: Small openings or spaces that form in enamel as it loses minerals during demineralization.
- **Fluoride**: A mineral that can be incorporated into enamel, making it more resistant to acid and caries.
- **Demineralization**: Loss of mineral content (mainly calcium and phosphate) from the enamel.

Zone 1: Translucent Zone

- **Advancing front**: The boundary where caries is actively progressing.
- **Prism boundaries**: The boundaries between individual enamel rods or prisms.
- **Structureless**: Lacking a defined, organized structure, often due to loss of mineral content.

Zone 2: Dark Zone

- **Excessive demineralization**: Greater than normal mineral loss in the enamel.
- **Remineralization**: The process of minerals being re-deposited into enamel from saliva, reversing some damage.

Zone 3: Body of the Lesion

- **Demineralization**: The loss of mineral content, leading to weakened enamel.
- **Pore volume**: The amount of space taken up by pores in the enamel, indicating the extent of demineralization.

Zone 4: Surface Zone

- **Polarizing light**: A technique used to view the structure of enamel, revealing details not visible with normal light.
- **Surface remineralization**: The process where minerals from saliva are deposited back into the surface enamel, repairing some of the damage.
- **Dentinal**: Pertaining to dentin, the tissue beneath the enamel.

Immunology of Dental Caries and Caries Activity Tests

Immunology of Dental Caries

- **Aciduric**: Refers to bacteria that can thrive in acidic environments, typically those found in the mouth during caries formation.
- **Organic acids**: Acids produced by bacteria during the fermentation of sugars, which contribute to demineralization of teeth.
- **Cariogenic**: Refers to substances or bacteria that cause or promote dental caries (tooth decay).

Caries Activity Tests

Lactobacillus Colony Test

- **Aciduric bacteria**: Bacteria that survive and grow in acidic conditions, such as those found in plaque during caries development.
- **Rogosa medium**: A type of growth medium used for cultivating aciduric bacteria, especially lactobacilli.

Snyder Test

- **Paraffin-stimulated saliva**: Saliva that is collected after chewing paraffin wax to stimulate saliva production.
- **Bromocresol green**: A pH indicator dye that changes color in response to pH changes, used to detect acid production.
- **Acidic by-products**: Products produced by bacteria from the metabolism of sugars that lower the pH in the mouth, leading to enamel demineralization.
- **pH range**: The scale used to measure acidity or alkalinity, where a lower value indicates greater acidity.

Swab Test

- **Oral flora**: The community of microorganisms that live in the mouth, including bacteria, fungi, and viruses.
- **Color comparator**: A tool used to compare the color change in a sample to a standard color scale to estimate pH changes.

Salivary S. mutans Level Test

- **Mitis Salivarius Agar (MSA)**: A special agar medium used to grow Streptococcus mutans, a bacterium associated with caries.
- **S. mutans**: A type of bacteria that plays a major role in the formation of dental caries.

S. mutans Dip Slide Method

- **Modified Mitis Salivarius Agar**: An altered version of MSA used to improve the growth of specific bacteria, such as S. mutans.
- **Colony density**: The concentration or number of bacterial colonies growing on an agar plate.

Buffer Capacity Test

- **Buffering capacity**: The ability of a solution (in this case, saliva) to resist changes in pH when acids are added.
- **pH meter**: A device used to measure the acidity or alkalinity of a solution.
- **Color indicators**: Substances that change color at different pH levels, used to assess acidity.

Enamel Solubility Test

- **Enamel solubility**: The degree to which enamel dissolves in acidic conditions.
- **Powdered enamel**: Ground or powdered enamel used in the test to assess how much of it dissolves when exposed to acidic saliva.

Salivary Reductase Test

- **Reductase enzyme**: An enzyme that catalyzes reactions where electrons are transferred from one molecule to another, typically reducing a compound.
- **Diazoresorcinol**: A chemical compound used as an indicator dye in the test, which changes color based on the activity of reductase enzymes.

Role of Bacterial Plaque in Dental Caries

Bacterial Plaque

- **Cariogenic microorganisms**: Bacteria that promote the development of dental caries by fermenting sugars and producing acids.
- **Fermentable substrates**: Carbohydrates and other organic compounds that can be broken down (fermented) by bacteria to produce acid.
- **Acquired pellicle**: A thin, protein-rich layer that forms on the surface of teeth from components in saliva. It serves as a foundation for bacterial attachment and plaque formation.
- **Glycoprotein**: A molecule made up of both protein and carbohydrate components, which forms part of the acquired pellicle.

Plaque's Role in Caries Development

- **Polysaccharides**: Complex carbohydrates stored by bacteria, which can be used as a reserve energy source, especially when dietary sugars are unavailable.
- **Buffering action**: The process by which saliva neutralizes acids in the mouth, helping to maintain a stable pH level.
- **Neutralization**: The process of reducing acidity, often by counteracting acids with alkaline substances such as those in saliva.

- **Retaining acids**: The ability of plaque to hold onto acids produced by bacteria, which increases the risk of demineralization of the enamel.
- **Barrier to salivary buffers**: The ability of plaque to prevent the neutralizing effects of saliva on acids in the mouth, allowing acids to persist longer on the tooth surface.

Sequelae of Dental Caries

Pulpal Involvement

- **Pulpitis**: Inflammation of the dental pulp, the innermost part of the tooth containing nerves and blood vessels.
- **Reversible Pulpitis**: A mild form of pulpitis where the inflammation can heal with appropriate treatment.
- **Irreversible Pulpitis**: Severe inflammation of the pulp, causing persistent pain and often requiring more invasive treatment.
- **Pulp Necrosis**: The death of the pulp tissue, resulting in the loss of tooth vitality.

Periapical Pathologies

- **Periapical Abscess**: A localized collection of pus at the tip of the tooth root caused by infection.
- **Granuloma**: A small area of chronic inflammation, often formed in response to infection, containing a mass of tissue.
- **Radicular Cyst**: A cyst located at the root of the tooth, often filled with fluid, and lined by epithelial cells.

Spread of Infection

- **Osteomyelitis**: An infection of the bone, which can occur when the infection from a tooth spreads to the jaw.
- **Cellulitis**: A skin infection that spreads rapidly in the soft tissue, leading to redness and swelling.
- **Ludwig's Angina**: A severe, life-threatening soft tissue infection in the neck and jaw, usually caused by infections in the mouth.

Tooth Loss

- **Functional issues**: Problems related to the tooth's ability to perform its normal function (e.g., chewing).
- **Aesthetic issues**: Problems related to the appearance of the tooth.

Systemic Complications

- **Bacteremia**: The presence of bacteria in the bloodstream, which can lead to other infections in the body.
- **Endocarditis**: Inflammation of the heart’s inner lining, usually caused by an infection that spreads from other parts of the body, like the mouth.